

Università degli Studi di Verona
Laurea Magistrale in Ingegneria e Scienze Informatiche

Quantum Computing — AA 2025–2026
Exam Project P2

Quantum Phase Estimation

Eigenphase of the controlled-phase gate CR_ϕ on $|11\rangle$ ■

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Implementation	Qiskit 2.x + Aer simulator
Language	Python 3

The circuit is implemented in Qiskit and demonstrated on two instances, as required by the exam project list.

1. Task

The primary function of Quantum Phase Estimation (QPE) is to find an approximate value of an eigenvalue phase of a unitary matrix U : a real number $0 < \vartheta < 1$ such that

$$U |u\rangle = e^{2\pi i \vartheta} |u\rangle$$

for some eigenvector $|u\rangle$ of U . The assigned instance is the controlled-phase gate $U = CR_\phi$ with eigenvector $|11\rangle$. The hint in the project list is to use $\vartheta = \phi / 2\pi$.

2. The unitary and its eigenphase

On two qubits the controlled-phase gate CR_ϕ leaves $|00\rangle$, $|01\rangle$ and $|10\rangle$ unchanged and multiplies $|11\rangle$ by a phase:

$$CR_\phi |11\rangle = e^{i\phi} |11\rangle$$

Writing the same eigenvalue in the QPE convention $e^{2\pi i \vartheta}$ gives $e^{i\phi} = e^{2\pi i \vartheta}$, hence

$$\vartheta = \phi / 2\pi, \text{ with } 0 < \vartheta < 1$$

This is exactly the hint in the exam brief. Powers of the gate are elementary: $(CR_\phi)^{2^k} = CR_{\phi \cdot 2^k}$, because $(e^{i\phi})^{2^k} = e^{i\phi \cdot 2^k}$.

3. The QPE circuit

Let $t = 3$ be the number of phase qubits. The algorithm is the standard one:

1. Prepare the eigenstate $|u\rangle = |11\rangle$ with two X gates on the target register.
2. Apply $H^{\otimes t}$ to the phase register, producing a uniform superposition.
3. For each phase qubit $k = 0, \dots, t-1$, apply the controlled operation U^{2^k} . Because $U |11\rangle = e^{i\phi} |11\rangle$, the target is unchanged and the phase $e^{i\phi \cdot 2^k}$ is kicked back onto qubit k (phase kick-back, as in the Deutsch lecture).
4. Apply the inverse quantum Fourier transform to the phase register.
5. Measure the phase register. If $\vartheta = m / 2^t$ for an integer m , the outcome is m with probability 1.

After the controlled powers the phase register is in the Fourier state

$$|\phi\rangle = 2^{-t/2} \sum_{k=0}^{2^t-1} e^{2\pi i \vartheta k} |k\rangle$$

The inverse QFT maps this state to $|m\rangle$ when $\vartheta = m / 2^t$. In the implementation, qubit 0 is the least significant bit: bit-reversal swaps are applied first, then Hadamard gates and negative controlled-phase gates from qubit 0 upward. This is the adjoint of the standard QFT that ends with a bit reversal.

In Qiskit, controlled- $CR_{\phi \cdot 2^k}$ is the multi-controlled phase gate $\text{mcp}(\phi \cdot 2^k)$ with controls on phase qubit k and on the first target qubit, and the phase applied to the second target qubit. The gate fires only on $|11\rangle$ of the target, which is the assigned eigenvector.

4. Chosen instances

With $t = 3$ qubits the phase grid is multiples of $1/8$. Two angles are chosen so that ϑ lies exactly on that grid. The circuit then succeeds with certainty, which makes the demonstration unambiguous.

ϕ	$\vartheta = \phi / 2\pi$	Binary ($t = 3$)	Integer m	Expected bitstring
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$\pi/2$	$1/4 = 0.250$	0.010	2	010
$\pi/4$	$1/8 = 0.125$	0.001	1	001

If ϑ were not a multiple of $1/2^t$, the measurement would spread around the closest integers and one would report the most frequent outcome. That case is not needed for the assigned demonstration.

5. Experimental results

The file `qpe.py` builds the circuit, transpiles it, and runs it on Aer with 2048 shots.

ϕ	True ϑ	Measured	Estimate	Shots
$\pi/2$	0.2500	010 = 2/8	0.2500	2048 / 2048
$\pi/4$	0.1250	001 = 1/8	0.1250	2048 / 2048

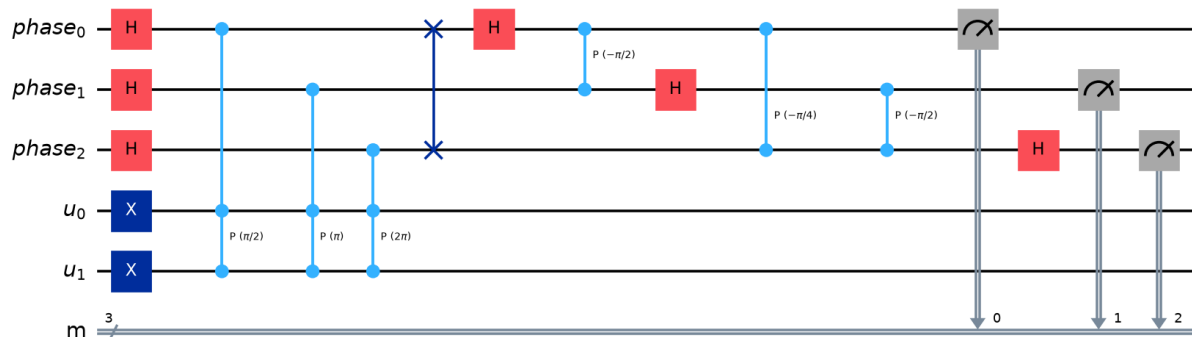


Figure 1. QPE circuit for $\phi = \pi/2$. The target is prepared in $|11\rangle$; each phase qubit controls a multi-controlled phase $\phi \cdot 2^k$; the inverse QFT is applied before measurement.

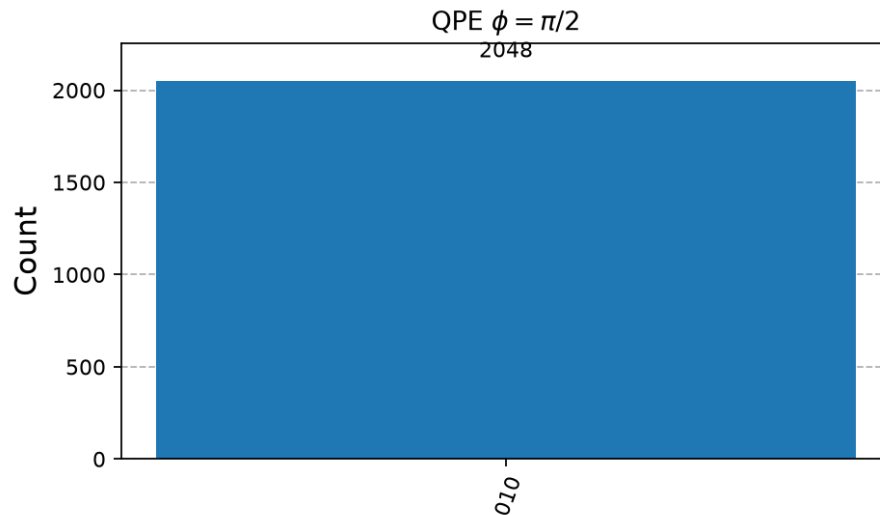


Figure 2. Histogram for $\phi = \pi/2$. Every shot returns 010, so $\vartheta = 2/8 = 1/4$.

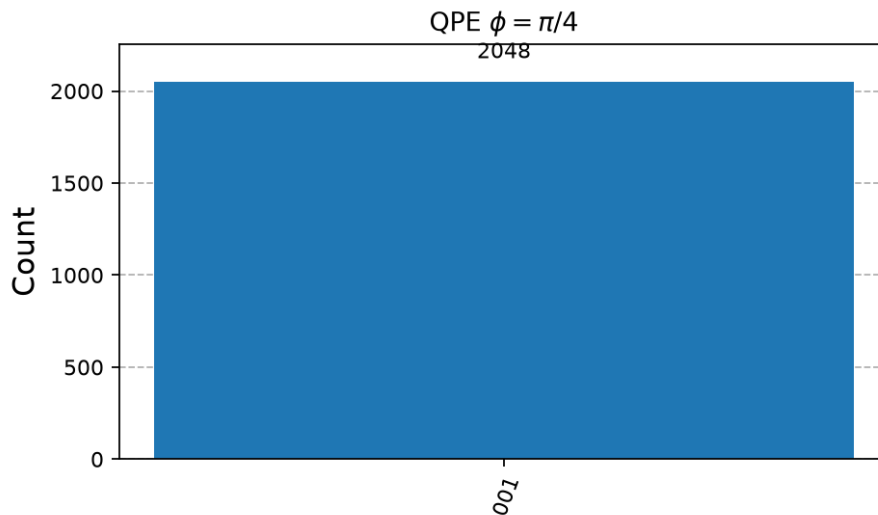


Figure 3. Histogram for $\phi = \pi/4$. Every shot returns 001, so $\vartheta = 1/8$.

6. How to run the demonstration

From the folder `p2-quantum-phase-estimation`:

```
pip install -r requirements.txt
python qpe.py
```

The Jupyter notebook `P2_Demo.ipynb` is the Qiskit demo. It states the eigenvalue relation, draws the circuit, runs both instances and plots the histograms.

7. Scope and limitations

The demonstration uses a noiseless simulator and an eigenphase that is exactly representable with 3 bits, as suggested by the hint $\vartheta = \phi / 2\pi$. QPE is the same subroutine that later appears in quantum counting (P3) and in Shor’s algorithm; those applications are outside this project. No hardware backend was used.

8. Conclusions

Quantum Phase Estimation was implemented in Qiskit for the controlled-phase gate CR_ϕ on the eigenvector $|11\rangle$. For $\phi = \pi/2$ and $\phi = \pi/4$ the circuit recovers $\vartheta = \phi / 2\pi$ on every shot, which matches the eigenphase relation required by the exam project.

References

- A. Di Pierro, *Quantum Computing*, lecture notes, University of Verona, AA 2025–2026. Project list: P2 Quantum Phase Estimation. Lectures on Deutsch (phase kick-back) and the circuit model.
- M. A. Nielsen and I. L. Chuang, *Quantum Computation and Quantum Information*, Cambridge University Press, Chapter 5 (quantum Fourier transform and phase estimation).
- P. Kaye, R. Laflamme and M. Mosca, *An Introduction to Quantum Computing*, Oxford University Press.
- Qiskit textbook, “Quantum Phase Estimation”. <https://qiskit.org/textbook>